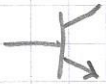
SINGLE BIPOLAR TRANSISTOR AMPLIFIER TYPES



① - COMMON EMITTER

② - COMMON COLLECTOR

③ - COMMON BASE

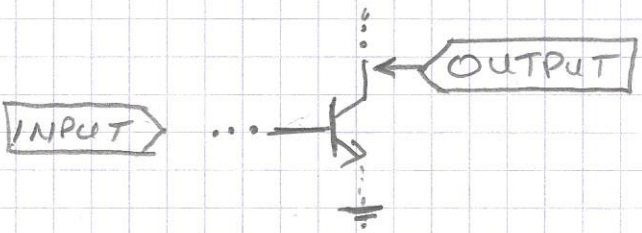
- DEFINITION
- CONFIGURATION
- BASIC CHARACTERISTICS
- TYPICAL USES

W2AEW

W2AEW

① COMMON EMITTER

- MOST COMMON TYPE OF G.P. AMPLIFIER

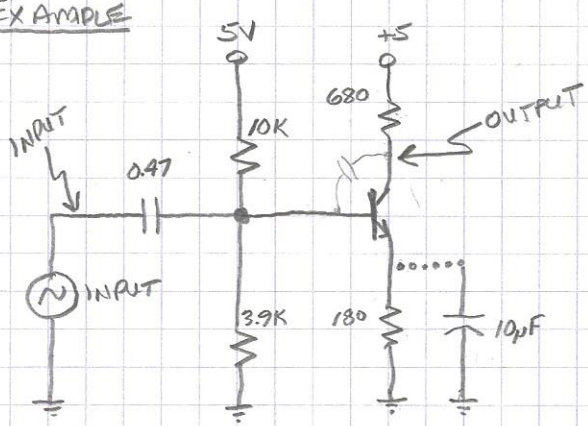


- INPUT SIGNAL CONNECTED TO BASE
- OUTPUT TAKEN FROM COLLECTOR

FEATURES

- MODERATE / HIGH INPUT IMPEDANCE
- MODERATE OUTPUT IMPEDANCE
- HIGH VOLTAGE GAIN
- HIGH CURRENT GAIN

EXAMPLE



- OUTPUT IS INVERTED
- MOST COMMON AMPLIFIER FOR MANY APPLICATIONS

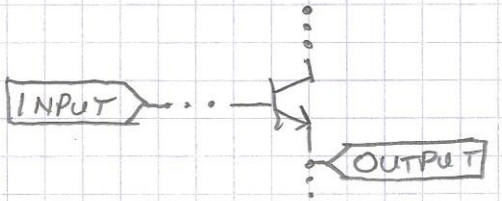
$$r_e = \frac{1}{g_m} = \frac{V_T}{I_c} \approx \frac{26mV}{I_c}$$

$$A_V \approx \frac{R_C}{r_e + R_E} \quad (g_m R_C)$$

W2AEW

② Common Collector

- Also called Emitter Follower

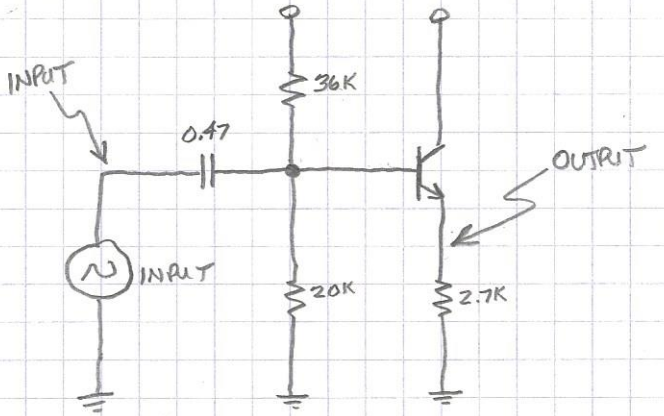


- Input connected to Base
- Output from Emitter

Features

- Moderate / High input impedance
- Low output impedance
- Low (unity) voltage gain
- High current gain

Example



- Most often used as a **Buffer**
- Easily drives various loads

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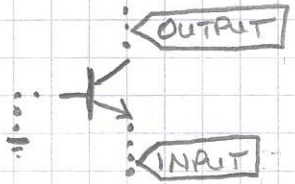
③ COMMON BASE

- LEAST COMMON, GOOD FOR RF, HIGH FREQUENCY

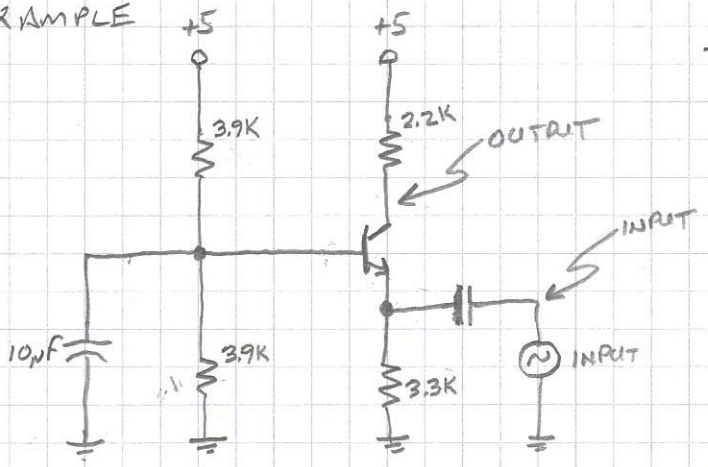


- INPUT CONNECTED TO EMITTER
- OUTPUT TAKEN FROM COLLECTOR

FEATURES

- LOW INPUT IMPEDANCE
- MODERATE / HIGH OUTPUT IMPEDANCE
- HIGH VOLTAGE GAIN
- ≈ UNITY CURRENT GAIN

EXAMPLE



- RF AMPLIFIERS